Prompt Engineering

Session 3 - Ethical & Responsible AI

0 | Quick Recap & Agenda

Session 4 - Group project assessment

Part 1 — Final Preparation (~1 hour) Use this time to finalize your group presentation.

Part 2 — Group Presentations (~2 hours) Each group: 10-minute presentation + 5-minute Q&A.

Session 2 — Key Takeaways

- **Be clear and direct:** explicit instructions beat vague requests. Add context and motivation behind your rules.
- **Structure with XML tags:** use tags to organize complex prompts with documents, examples, and variables.
- **System prompts:** the hidden layer that shapes every response. Define persona, constraints, and output format.
- **Chain of Thought:** CoT and reasoning modes dramatically improve complex reasoning tasks.
- **Meta-prompting:** use AI to help you write better prompts. Iterate collaboratively with the model.
- **Prompt chaining:** break complex tasks into focused steps. Use the self-correction pattern for reliability.

Today's Agenda

Part 1 - Why Ethics in AI Matters

~20 min

- Real-world AI failures and their consequences
- Bias, deepfakes, privacy, and environmental impact

Part 2 - The EU AI Act Explained

~25 min

- Risk-based framework, categories, obligations, and penalties
- Concrete examples for each risk level

Part 3 - Responsible AI in Practice

~15 min

- Practical ethical checklist, workshop, and group discussion

Part 4 - Group Project

~2 hours

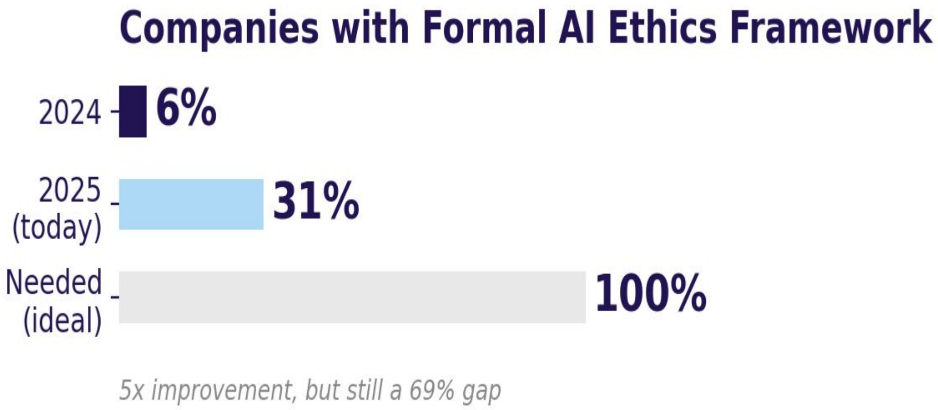
- Prepare for your final session group project

1 | Why ethics in AI Matters

The Power and the Responsibility

The Power and the Responsibility

We are only at the very beginning. Here is where we stand today vs. what is coming.



The question is not
IF we should use AI…
but HOW we use it responsibly.

Sources: Gartner, PwC, Deloitte 2024, DataReportal 2025, IDC 2025

Real-World AI Failures

- **Amazon Recruiting Tool (2018):** Amazon built an AI to screen resumes. It penalized CVs containing the word "women's".. The system was trained on 10 years of male-dominated hiring data and learned that male = better. Amazon scrapped the tool.
- **COMPAS Algorithm (USA):** used in US courts to predict recidivism. ProPublica found it was twice as likely to falsely flag Black defendants as future criminals compared to white defendants. Judges relied on it for bail and sentencing decisions.
- **Microsoft Tay (2016):** a Twitter chatbot that learned from user interactions. Within 24 hours, it was posting racist and offensive tweets. Microsoft shut it down in less than a day. It showed how AI can amplify the worst of human behavior.

Bias in AI Systems

AI systems learn from data. If the data reflects historical biases, the AI will reproduce and amplify them.

- **Healthcare bias:** a widely-used algorithm in US hospitals was found to systematically deprioritize Black patients for extra care. It used healthcare spending as a proxy for health needs, but Black patients historically had less access to care and thus lower spending.
- **Facial recognition:** studies by MIT showed commercial facial recognition systems had error rates of up to 34.7% for dark-skinned women, compared to 0.8% for light-skinned men.
- **Language models:** GPT and other LLMs have been shown to associate certain professions with specific genders (e.g. "nurse" = female, "CEO" = male) and to generate stereotypical content about minority groups.

Deepfakes & Misinformation

- **Political deepfakes:** in February 2024, a deepfake video manipulated a France 24 news bulletin, using AI to clone the presenter's voice and face to falsely claim that President Macron had canceled a trip to Ukraine over an assassination plot. The video was traced to pro-Russian propaganda networks and went viral before being debunked.

- **Financial fraud:** a Hong Kong finance worker was tricked into transferring $25M after a video call with deepfakes of his colleagues and CFO. The entire call was AI-generated.

- **Academic fraud:** AI-generated scientific papers with fabricated data and fake references are flooding academic journals. Retraction Watch reported a 10x increase in 2024.

Deepfakes - What do you think?

Privacy Concerns

- **Clearview AI:** scraped over 30 billion facial images from the internet (social media, news sites) without consent. Used by police in 27+ countries. Fined over 70M euros in the EU.
- **Samsung data leak (2023):** Samsung employees accidentally pasted confidential semiconductor source code into ChatGPT. The data became part of the training context. Samsung banned ChatGPT internally.
- **Italy vs. ChatGPT (2023):** Italy temporarily banned ChatGPT over GDPR concerns: no age verification, no legal basis for data collection, no way for users to correct inaccurate outputs about themselves.
- **Training data lawsuits:** The New York Times, Getty Images, and thousands of authors have filed lawsuits against AI companies for using copyrighted content without permission to train their models.

Environmental Impact of AI

- **Training GPT-4:** estimated to have consumed ~50 GWh of energy and produced ~12,000 tons of CO_2, equivalent to the annual emissions of ~2,600 cars.
- **Water consumption:** Microsoft reported a 34% increase in water consumption in 2022, largely attributed to AI data center cooling. Training GPT-3 alone consumed ~700,000 liters of fresh water.
- **Daily usage:** a single ChatGPT query uses roughly 10x more energy than a Google search. With 200M+ weekly active users, the cumulative impact is massive.
- **E-waste:** AI requires specialized hardware (GPUs, TPUs) with short lifecycles. Data centers replace hardware every 3-5 years, contributing to growing electronic waste.

2 | The EU AI Act Explained

What is the EU AI Act?

The EU AI Act is the world's first comprehensive legal framework for artificial intelligence. Adopted in March 2024 by the European Parliament, it sets rules for how AI systems can be developed, deployed, and used within the European Union.

- **Scope:** applies to any AI system placed on the EU market or whose output is used in the EU, regardless of where the provider is based. This means US and Chinese AI companies are also affected.
- **Approach:** uses a risk-based classification system. The higher the risk, the stricter the requirements.
- **Goal:** protect fundamental rights (health, safety, non-discrimination) while fostering innovation and competitiveness in Europe.
- **Inspiration:** follows the same regulatory philosophy as GDPR. Just as GDPR became the global standard for data privacy, the AI Act aims to set the global standard for AI governance.

AI Act Timeline

April 2021	European Commission publishes the first proposal for the AI Act.
December 2023	Political agreement reached between EU Parliament and Council after marathon negotiations.
March 2024	European Parliament formally adopts the AI Act with overwhelming majority (523 votes in favor).
August 2024	AI Act enters into force (published in Official Journal).
February 2025	Bans on unacceptable-risk AI systems take effect.
August 2025	Obligations for general-purpose AI (GPAI) models apply.
August 2027	Full enforcement: all high-risk AI obligations apply. Penalties become enforceable.

The Risk-Based Approach

The AI Act classifies AI systems into 4 risk categories. The higher the risk to health, safety, or fundamental rights, the stricter the rules.

UNACCEPTABLE RISK

Banned

HIGH RISK

Strictly regulated

LIMITED RISK

Transparency obligations

MINIMAL RISK

Free use

Unacceptable Risk — Banned Practices

These AI applications are completely prohibited in the EU. No exceptions.

- **Social scoring by governments:** AI systems that rate citizens based on their behavior or personal characteristics (like China's Social Credit System). A government cannot deny you housing based on an AI score of your social media activity.
- **Subliminal manipulation:** AI that exploits psychological vulnerabilities to distort behavior in harmful ways. Example: an AI toy that uses voice commands to encourage dangerous behavior in children.
- **Real-time biometric surveillance:** mass facial recognition in public spaces by law enforcement (with narrow exceptions for terrorism, kidnapping). A city cannot install AI cameras that identify every person walking in the street.
- **Emotion recognition at work/school:** AI systems that infer emotions from facial expressions or voice in the workplace or educational institutions. Your employer cannot use AI to detect if you are "disengaged" during meetings.

High Risk — Strictly Regulated

These systems impact fundamental rights and must meet strict requirements before deployment.

- **Recruitment AI:** any AI that screens CVs, ranks candidates, or makes hiring recommendations. Example: HireVue's video analysis tool that scores candidates based on facial expressions and word choice.
- **Credit scoring:** AI systems that decide whether you get a loan or a mortgage. Banks must ensure the AI does not discriminate based on ethnicity, gender, or zip code.
- **Education:** AI that determines exam grades or school admissions. Example: the UK's A-level algorithm in 2020 that downgraded students from disadvantaged areas.
- **Law enforcement:** predictive policing tools (except banned real-time surveillance). AI that flags suspects or assesses risk levels must be transparent and auditable.
- **Medical devices:** AI-powered diagnostic tools (radiology AI, pathology assistants) must meet the same standards as traditional medical devices.

High-Risk AI: What Companies Must Do

- Data governance: training data must be relevant, representative, and free of errors. Providers must document data sources and preprocessing steps.
- Technical documentation: detailed description of the system's design, development, testing, and intended purpose. Must be kept up to date.
- Record-keeping (logging): the AI must automatically log its activities so that operations can be traced and audited after the fact.
- Transparency: users must be clearly informed that they are interacting with an AI system. Instructions for use must be provided.
- Human oversight: a qualified human must be able to understand, monitor, and override the AI's decisions. No fully autonomous high-risk AI.
- Accuracy, robustness, cybersecurity: the system must perform reliably and be resilient to adversarial attacks and data manipulation.

Limited Risk — Transparency Obligations

These systems pose moderate risks. The main obligation is transparency: users must know they are interacting with AI.

- **Chatbots:** any AI chatbot must clearly inform the user that they are talking to a machine, not a human. Example: customer service bots must display a disclaimer like "You are chatting with an AI assistant."
- **AI-generated content:** images, audio, video, or text generated by AI must be labeled as such. Example: an AI-generated news article must carry a visible label.
- **Deepfake disclosure:** any content that has been artificially manipulated (face-swapped video, cloned voice) must be marked. Example: a political ad using AI-altered footage must disclose this.
- **Emotion detection:** if an AI detects emotions (outside of banned contexts), it must inform the person being analyzed.

Minimal Risk — Free Use

The vast majority of AI systems fall into this category. No specific obligations, but voluntary codes of conduct are encouraged.

- Spam filters: your email spam filter uses AI to classify messages. No regulation needed.
- Video game AI: NPC behavior, difficulty adjustment, and procedural content generation are all minimal risk.
- Recommendation engines: Netflix suggesting your next show, Spotify creating playlists. These are considered minimal risk under the Act (though they raise ethical debates about filter bubbles).
- AI writing assistants: Grammarly, autocomplete features, and translation tools like DeepL fall here.
- Search engines: basic search algorithms and content ranking systems.

General-Purpose AI (GPAI) — Special Rules

The AI Act created a specific category for foundation models like GPT-5, Gemini, Claude… These are not task-specific, so they get their own rules.

- **All GPAI providers must:** publish detailed technical documentation, provide a summary of training data, comply with EU copyright law, and designate an EU representative.
- **Systemic risk models:** GPT-5, Gemini Ultra, Claude Opus fall here. They must additionally: perform adversarial testing (red-teaming), report serious incidents, ensure cybersecurity protections, and report energy consumption.
- **Open-source exception:** open-source GPAI models (like Llama, Mistral) have lighter obligations, unless they pose systemic risk.
- **Why it matters:** these rules directly affect the tools you use every day. ChatGPT, Claude, Gemini all must comply.

Penalties & Enforcement

The AI Act has teeth. Penalties are designed to be dissuasive, following the GDPR model.

35M euros

or 7% of global turnover

For deploying banned (unacceptable risk) AI

15M euros

or 3% of global turnover

For violating high-risk AI obligations

7.5M euros

or 1.5% of global turnover

For providing incorrect information to authorities

For context: 7% of Google's 2023 revenue would be ~$21 billion. These are not trivial fines.

Quick Quiz — Classify the Risk Level

For each scenario, decide: Unacceptable, High, Limited, or Minimal risk?

1. A city government installs AI cameras in every street to score citizens based on their social behavior.
2. A hospital uses an AI to assist radiologists in detecting lung cancer from X-rays.
3. A customer service website uses a chatbot to handle return requests.
4. Spotify uses AI to recommend songs you might like.
5. A bank uses AI to decide whether to approve mortgage applications.
6. A political party uses AI to generate campaign videos showing their opponent saying things they never said.
7. A school uses AI to monitor students' facial expressions during exams to detect cheating.

Quiz — Answers

1. **Social scoring by government:** UNACCEPTABLE RISK — explicitly banned. This is the Social Credit System scenario.
2. **AI-assisted radiology:** HIGH RISK — medical device AI requires full compliance: testing, documentation, human oversight.
3. **Customer service chatbot:** LIMITED RISK — must disclose it's an AI. No other major obligations.
4. **Spotify recommendations:** MINIMAL RISK — no specific obligations. Standard consumer AI application.
5. **Bank mortgage AI:** HIGH RISK — credit scoring AI directly affects fundamental rights (access to housing).
6. **AI-generated political deepfake:** LIMITED RISK (must be labeled) + potentially UNACCEPTABLE if used for subliminal manipulation
7. **Emotion detection in school:** UNACCEPTABLE RISK — emotion recognition in educational institutions is explicitly banned.

3 Responsible AI in practice

Transparency: Always Disclose AI Use

- **In professional work:** if you used AI to draft a report, write an email, or generate code, disclose it. Clients and colleagues deserve to know. Example: "This market analysis was produced with the assistance of Claude AI and reviewed by the team."

- **In academic work:** most universities now have AI use policies. Eg Albert School, you should always declare when and how you used AI tools. Transparency builds trust and credibility.

- **In content creation:** AI-generated images, articles, and videos should be labeled. The EU AI Act requires this for limited-risk systems, but ethical practice goes beyond legal minimums.

- **In decision-making:** if AI influenced a hiring decision, a medical recommendation, or a legal assessment, the affected person has a right to know.

Checking for Bias — Practical Steps

- **Question your data:** who collected it? Who is represented? Who is missing? If your training dataset contains mostly English-language text from Western sources, it will have blind spots for other cultures.
- **Test across demographics:** run your AI output through different scenarios. Does a resume screener score differently based on names? Does a content generator produce stereotypes about certain groups?
- **Use diverse teams:** homogeneous teams build biased products. Include diverse perspectives in AI design, testing, and deployment decisions.
- **Monitor continuously:** bias can emerge over time as the world changes. A model trained in 2024 may become biased by 2026 if it is not regularly evaluated.
- **Document and report:** keep records of bias testing. If you discover bias, report it and take corrective action. Hiding known bias is both unethical and increasingly illegal.

AI and Academic Integrity : Knowledge vs Understanding

As students, how should you use AI tools ethically?

- **AI as a tutor, not a ghostwriter:** use AI to explain concepts, check your reasoning, or brainstorm ideas. Do not use it to write your assignments for you. The learning happens in the doing.
- **Always disclose:** if your professor allows AI use, state clearly how you used it. "I used ChatGPT to generate an initial outline, then rewrote and expanded every section myself."
- **Verify everything:** AI hallucinations are real. It can generate fake citations, invent statistics, and state incorrect facts with complete confidence. Always fact-check.
- **Develop your own skills:** if you rely on AI for everything now, you will not develop the critical thinking, writing, and analysis skills that employers actually value.

Workshop — Ethical Dilemma Analysis (15')

In your project groups, analyze ONE of these scenarios. Present your analysis in 3 minutes.

- Scenario A: A startup offers an AI that analyzes job candidates' social media profiles to predict their "cultural fit." Legal? Ethical? What could go wrong?
- Scenario B: A university deploys AI to monitor online exam sessions via webcam, flagging "suspicious behavior" (looking away, lip movements). Is this acceptable? What are the risks?
- Scenario C: A news agency uses AI to generate 50 articles per day, publishing them under human-sounding bylines with no AI disclosure. Is this ethical? Should it be legal?
- Scenario D: A government uses AI to process asylum applications, recommending approval or rejection based on applicant profiles. What safeguards are needed?

Workshop — Analysis Framework

Use this framework to structure your group analysis:

1. **Classification:** What risk level does this scenario fall under in the EU AI Act? Justify your answer.
2. **Stakeholder analysis:** Who benefits? Who is harmed? Who has no voice in the process
3. **Ethical assessment:** Is this application ethical, even if it is technically legal? What principles apply (fairness, transparency, autonomy, privacy)?
4. **Practical safeguards:** What concrete measures would you recommend to make this AI system more responsible?
5. **Your verdict:** Should this AI application be allowed, modified, or banned? Defend your position.

Session 3 — Key Takeaways

- **AI has real consequences:** biased algorithms, deepfakes, privacy violations, and environmental costs are not hypothetical. They are happening right now.
- **The EU AI Act is landmark legislation:** it creates a risk-based framework with real penalties. It applies to any AI used in Europe, regardless of where it was built.
- **4 risk levels:** unacceptable (banned), high (strictly regulated), limited (transparency), minimal (free use). Know where your AI application falls.
- **GPAI models have special rules:** ChatGPT, Claude, Gemini must comply with documentation, copyright, and transparency requirements.
- **Ethics goes beyond the law:** legal compliance is the floor, not the ceiling. Always disclose AI use, check for bias, verify outputs, and consider the broader impact.
- **Your checklist:** transparency, accuracy, privacy, fairness, human oversight, skill development, environmental awareness, and public accountability.

Looking Ahead — What's Next

- **August 2026:** full enforcement of the EU AI Act. All high-risk obligations kick in. The regulatory landscape will shift dramatically.
- **AI literacy is becoming essential:** understanding not just how to use AI, but how to use it responsibly, will be a key differentiator in the job market.
- **Your role as future professionals:** you will be the generation that implements AI governance in companies. Understanding the AI Act now gives you a competitive edge.

Thanks!

Questions?